

RBSE 2020

Grade 12

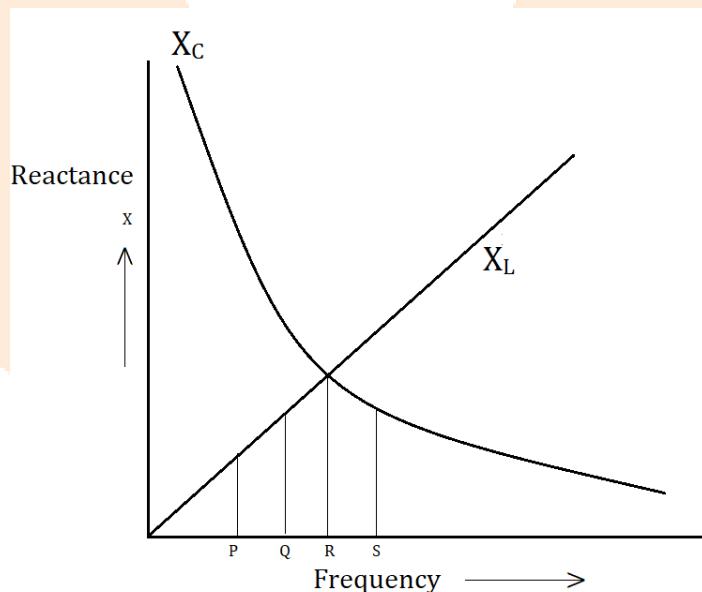
Physics

Time: 3 ¼ Hours

Max. Marks: 56

SECTION – A

1. write the definition of dielectric constant of a medium. **1**
2. Why do we get balancing point in the middle of the meter bridge generally? **1**
3. Write the name of any two rules to find the direction of magnetic field. $\frac{1}{2} + \frac{1}{2} =$ **1**
4. What is diamagnetic substances? **1**
5. define magnetic flux. **1**
6. In the given figure, write a point showing the resonant state. **1**



7. What do you understand by scattering of light? 1
8. What was the main objective of Davisson and Germer experiment. 1
9. Write Heisenberg's uncertainty principle. 1
10. Write the function of coolant used in nuclear reactor. 1
11. Draw symbolic diagram of XOR gate. 1
12. Draw block diagram of a communication system. 1
13. What is nanotechnology? 1

SECTION-B

14. An electron is moving around an infinite charge in circular path of diameter 0.30 m. If linear charge density is 10^{-6} C/m, then calculate the speed of an electron.
($m_e = 9.0 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C) 2
15. Three resistors are connected in parallel. The electric current 7.5 A is flowing in this combination from a battery of emf 30 V. If the values of two resistors are $10\ \Omega$ and $12\ \Omega$, then find the value of third resistor. 2
16. (a) Write Kirchhoff's Second law.
(b) Draw a circuit diagram for comparison of emf of two primary cells with the help of potentiometer. 1+1=2
17. (a) The magnetic moment of two bar magnets are $\vec{M}_1$ and $\vec{M}_2$. If angle between them is θ , then write the value of resultant magnetic moment.

OR

(b) Soft iron used in making electro-magnets. Why?

1+1=2

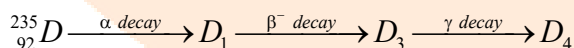
18. Focal length of a convex lens in air is 25 cm. If it is immersed in water, then calculate the focal length of the lens. $\left(n_w = \frac{4}{3}, n_g = \frac{3}{2}\right)$. **2**
19. Write any two differences between diffraction and interference of light. **1+1=2**
20. Why photo-electric effect cannot be explained on the basic of wave theory of light? Write any two reasons. **1+1=2**
21. The power factor of a coil is 0.707 at frequency 50 Hz, If frequency is doubled, then calculate the power factor of coil. **2**

OR

The voltage and current in a.c. circuit are $V = 50 \sin 314 t$ V and $I = 10 \sin\left(314t + \frac{\pi}{4}\right)$ A respectively Calculate: **1+1=2**

(i) Wattless current and (ii) Root mean square voltage.

22. A radioactive nucleus 'D' is decaying in the following way:



Determine the atomic number and mass number of D_4

1+1=2

23. What is intrinsic semiconductor and extrinsic semiconductor? Write an example of each semiconductor

1+1=2

24. The maximum and minimum amplitude for an amplitude modulated wave are 10V and 2V respectively. Calculate the modulation index. **2**
25. Obtain an expression for the induced emf $e = e_0 \sin \omega t$ generated due to the rotational motion of the rectangular coil in the uniform magnetic field. Draw a graph between induced emf 'e' and time 't' **2 + 1 = 3**
26. Explain Bohr's second postulate with the help of De-Broglie's hypothesis. Draw a Stationary wave model of an electron for orbit $n=3$ **2 + 1 = 3**
27. What is P-N junction diode? Draw a circuit diagram of the Zener diode as a voltage regulator (Voltage stabilizer). Write its working also. **1+1+1=3**

OR

What is amplification? Draw a circuit diagram of NPN transistor as a common emitter amplifier. Write its working also. **1+1+1=3**

28. Define electric potential. Obtain an expression of electric potential due to an electric dipole at any point (r, θ). Draw necessary diagram. **1 + 2 + 1 = 4**

OR

What is capacitor? Derive an expression of capacitance of the spherical capacitor. Draw necessary diagram.

1 + 2 + 1 = 4

29. Describe the construction of suspended coil galvanometer, draw labelled diagram of it. Prove that current flowing in the coil of galvanometer is directly proportional to the deflection generated in the coil.

$$1 + 2 + 1 = 4$$

OR

What is Helmholtz coils? Prove that the value of a uniform magnetic field obtained in The Helmholtz coil is 1.432 time the magnetic field generated by each coil at its Centre.

$$1+2+1=4$$

30. What is Presbyopia? Describe the construction and working of reflecting telescope. Draw necessary ray diagram.

$$1 + 2 + 1 = 4$$

OR

What is wave front? Explain laws of refraction of light on the basis of Huygens wave theory. Draw necessary ray diagram.

$$1 + 2 + 1 = 4$$